

PLS: $\mathbf{t}_1 = \mathbf{X}\mathbf{w}_1$

Intuitive meaning

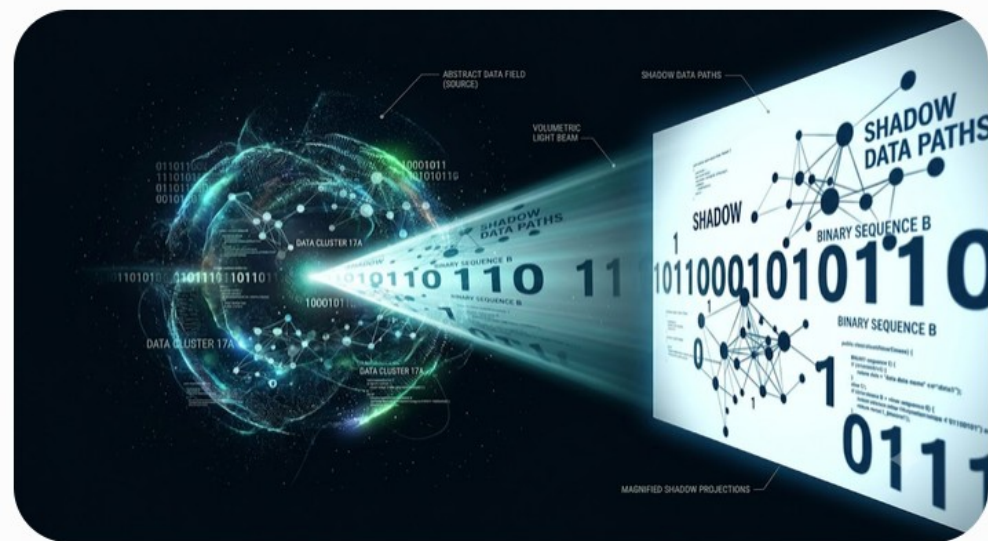
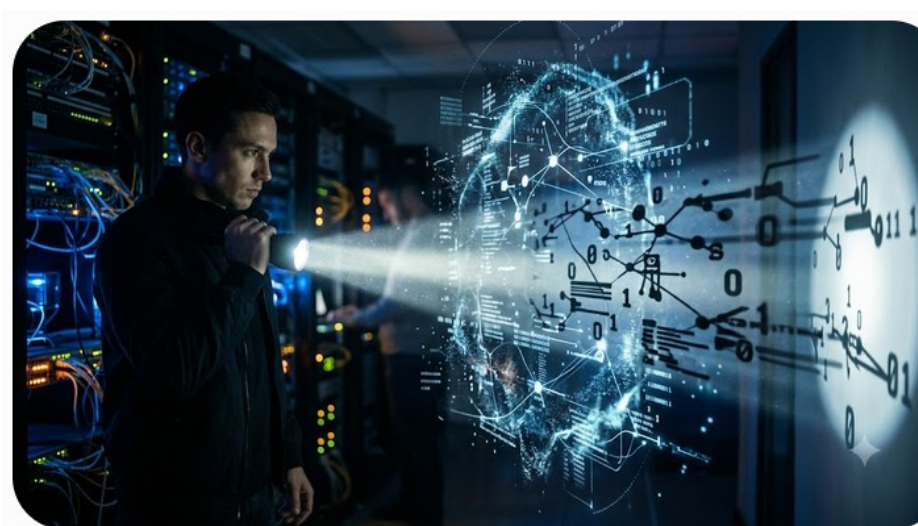
Think of the data matrix as

	X_1	X_2	$\dots$	X_p
Obs 1				
Obs 2				
$\vdots$				
Obs n				

- **Weight vector** $\mathbf{w}_1$: defines the **direction** in the p -dimensional variable space.
- **Score vector** $\mathbf{t}_1$: gives the **location (projection)** of each of the n observations along that direction.

An analogy is shining a flashlight onto the data cloud:

- the **weight vector** is the direction of the flashlight beam,
- the **score vector** contains the positions (shadows) of all observations on that beam.



Quantity	Dimension	Represents
$\mathbf{w}_1$	$p \times 1$	Direction (combination of variables) defining the first latent variable
$\mathbf{t}_1$	$n \times 1$	Coordinates (scores) of all observations after projection onto that direction

X and Y

- ❖ Let's consider the calculations associated with the first PLS factor.
- ❖ Let $\mathbf{X}$ be an $n \times p$ matrix whose columns are the p predictors and
- ❖ $\mathbf{Y}$ is an $n \times k$ matrix whose columns are the k responses

- $n = 10$ observations
- $p = 5$ predictor variables
- $k = 3$ response variables

Predictor Matrix $\mathbf{X}$ (10×5)

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & 6 & 5 & 3 \\ 3 & 5 & 7 & 6 & 4 \\ 4 & 7 & 9 & 8 & 5 \\ 5 & 8 & 10 & 9 & 6 \\ 6 & 9 & 12 & 11 & 8 \\ 7 & 11 & 13 & 12 & 9 \\ 8 & 12 & 15 & 14 & 10 \\ 9 & 14 & 17 & 15 & 12 \\ 10 & 15 & 18 & 17 & 13 \\ 11 & 17 & 20 & 18 & 15 \end{bmatrix}$$

Response Matrix $\mathbf{Y}$ (10×3)

$$\mathbf{Y} = \begin{bmatrix} 20 & 12 & 30 \\ 24 & 15 & 35 \\ 29 & 18 & 41 \\ 33 & 21 & 46 \\ 39 & 25 & 54 \\ 43 & 28 & 59 \\ 49 & 31 & 66 \\ 54 & 35 & 73 \\ 58 & 38 & 79 \\ 64 & 42 & 87 \end{bmatrix}$$

Validation with JMP

Model Coefficients for Original Data

Term	Y1		Y2		Y3	
Intercept	0.0000		0.0000		0.0000	
X1	0.5236		0.3382		0.7172	
X2	0.8125		0.5247		1.1128	
X3	0.9976		0.6443		1.3664	
X4	0.9075		0.5860		1.2428	
X5	0.6864		0.4432		0.9400	

>> Beta_P

Beta_P =

0.5236	0.3382	0.7172
0.8125	0.5247	1.1128
0.9976	0.6443	1.3664
0.9075	0.5860	1.2428
0.6864	0.4432	0.9400

Leave-One-Out Cross Validation With Me

Number of factors	Root Mean PRESS
0	48.229390
1	2.792076

RMSECV =

2.7921

X Score Y Score

9.400438	37.93491
11.58778	44.98066
15.28858	53.33225
17.47592	60.378
21.10648	71.14607
23.74631	78.19183
26.99462	87.84926
30.57231	97.31149
33.26502	105.112
36.8427	115.8801

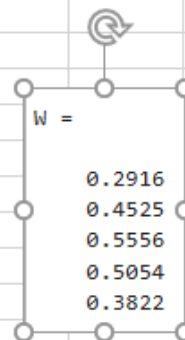
T =

9.4004
11.5878
15.2886
17.4759
21.1065
23.7463
26.9946
30.5723
33.2650
36.8427

U =

37.9349
44.9807
53.3323
60.3780
71.1461
78.1918
87.8493
97.3115
105.1120
115.8801

X Variables	X Weights 1
X1	0.2916184696
X2	0.4524980245
X3	0.5556001807
X4	0.5053737157
X5	0.3822465278



PLS: Theory

$$\mathbf{w}_1 = \begin{bmatrix} 0.245 \\ 0.408 \\ 0.612 \\ 0.490 \\ 0.408 \end{bmatrix}$$

$$\mathbf{t}_1 = \begin{bmatrix} -4.52 \\ -3.61 \\ -2.75 \\ -1.84 \\ -0.88 \\ 0.12 \\ 1.06 \\ 2.08 \\ 3.11 \\ 4.23 \end{bmatrix}$$

- ❖ The weight vector $\mathbf{w}_1$ ($p \times 1$) is used to weight each of the observations in $\mathbf{X}$ ($1 \times p$).
- ❖ Do it for all the n observations of $\mathbf{X}$ to get the Latent Vector $\mathbf{t}_1$ ($n \times 1$)
- ❖ The n weighted linear combinations of the entries in the columns of $\mathbf{X}$ are called X scores, denoted by the vector $\mathbf{t}_1$.
- ❖ The X scores are the entries of the vector $\mathbf{t}_1 = \mathbf{X}\mathbf{w}_1$ ($n \times p \times p \times 1$)
- ❖ Note that the size of the score vector, $\mathbf{t}_1$, is $n \times 1$

PLS:

$$t_1 = Xw_1$$

Weight vector w_1

$$w_1 = \begin{bmatrix} w_{11} \\ w_{21} \\ \vdots \\ w_{p1} \end{bmatrix}, \quad p \times 1$$

- Dimension: $p \times 1$
- One element per predictor variable.
- Each weight indicates how much that predictor contributes to forming the first latent variable.

If there are

- $p = 4$ variables,

then

$$w_1 = \begin{bmatrix} 0.48 \\ 0.72 \\ -0.39 \\ 0.31 \end{bmatrix}$$

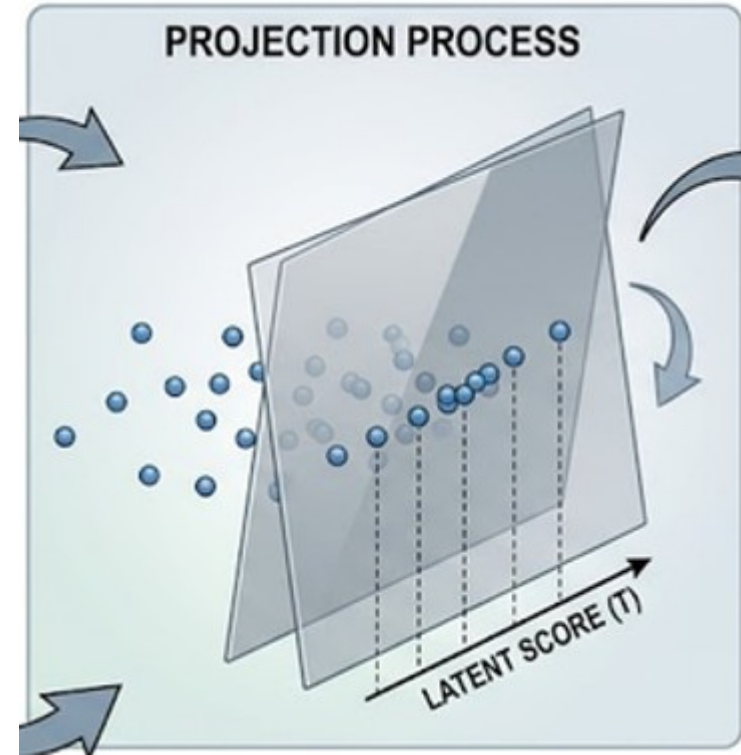
Here,

- Variable 1 contributes with weight 0.48,
- Variable 2 contributes most strongly (0.72),
- Variable 3 contributes negatively (-0.39),
- Variable 4 contributes moderately (0.31).



PLS: $t_1 = Xw_1$

- ❖ Values in the column of t_1 matrix in PLS are called as X scores
- ❖ Think of the vector w_1 as defining a linear transformation mapping ALL the p predictors to a **one dimensional subspace**
- ❖ With this interpretation, Xw_1 represents the mapping of the data to this one-dimensional



PLS: Theory

Technically, $\mathbf{t}_1$ is a *linear combination of the variables in $\mathbf{X}$* that has *maximum covariance*

with a *linear combination of the variables in $\mathbf{Y}$* ,

subject to normalizing constraints.

❖ the **COVARIANCE** between

$\mathbf{t}_1 (= \mathbf{X}\mathbf{w}_1)$ and $\mathbf{u}_1 (= \mathbf{Y}\mathbf{q}_1)$

is a **MAXIMUM**.

❖ The vector $\mathbf{q}_1$ is a $\mathbf{Y}$ weight vector

❖ The elements of the vector $\mathbf{u}_1$ are the $\mathbf{Y}$ scores

❖ For the First factor, we expect the X scores and the Y scores to be strongly correlated

```
>> Compare=[T U]
```

```
Compare =
```

9.4004	37.9349
11.5878	44.9807
15.2886	53.3323
17.4759	60.3780
21.1065	71.1461
23.7463	78.1918
26.9946	87.8493
30.5723	97.3115
33.2650	105.1120
36.8427	115.8801

```
>> corr(Compare)
```

```
ans =
```

1.0000	0.9998
0.9998	1.0000

PLS: Theory

- ❖ To obtain subsequent factors, first, we use all factors available to that point to predict BOTH X and Y
- ❖ In the NIPALS algorithm, the process of obtaining a new weight vector and defining new X scores is applied to the **residuals** from the predictive models for X and Y
- ❖ We say that X and Y are deflated into predictions + residuals and the process itself is called **deflation**
- ❖ This ensures that subsequent factors are independent of (**orthogonal** to) all previously extracted factors

PLS: Theory

❖ Suppose that 'a' factors are extracted. Then there are:

'a' weight vectors, $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_a$

'a' X-score vectors, $\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_a$

'a' Y-score vectors, $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_a$

❖ We can now define three matrices:

❖ $\mathbf{W}$ is the $p \times a$ matrix whose columns consist of the weight vectors;

❖ $\mathbf{T}$ and $\mathbf{U}$ are the $n \times a$ matrices whose columns consist of the X-score & Y-score vectors, respectively

```
num_components =
```

```
3
```

```
>> p
```

```
p =
```

```
5
```

```
>> n
```

```
n =
```

```
10
```

```
W =
```

0.2916	-0.4307	-0.5052
0.4525	-0.1913	0.0334
0.5556	0.5598	0.4341
0.5054	0.2757	-0.5745
0.3822	-0.6233	0.4745

```
T =
```

9.4004	1.2595	0.3779
11.5878	0.8520	0.2076
15.2886	1.0930	-0.0208
17.4759	0.6855	-0.1911
21.1065	0.4931	-0.0078
23.7463	-0.1055	-0.1590
26.9946	0.3266	-0.4065
30.5723	-0.3342	0.3543
33.2650	-0.4644	-0.3744
36.8427	-1.1252	0.3864

```
>> U
```

```
U =
```

37.9349	7.6281	1.1686
44.9807	7.4002	2.9914
53.3323	3.7297	-1.8642
60.3780	3.5018	-0.0414
71.1461	2.4192	-0.1251
78.1918	0.7652	1.2918
87.8493	-0.1521	-1.9190
97.3115	-2.3952	-0.6942
105.1120	-3.3353	-0.8873
115.8801	-4.2513	1.5764

'a' weight vectors,

$w_1, w_2, \dots, w_a$

'a' X-score vectors,

$t_1, t_2, \dots, t_a$

'a' Y-score vectors,

$u_1, u_2, \dots, u_a$

Philosophy of PLS2 Algorithm

Standard multiple regression, may **fail** if predictors are

- ❖ highly correlated (multicollinearity) or
- ❖ outnumber the observations,
- ❖ PLS2 avoids overfitting by doing dimensionality reduction in 2 ways

1. **Latent Variable Extraction:** It projects both the **X** and **Y** matrices into a new, smaller space of "latent variables".

2. **Maximizing Covariance:** Instead of just finding components that explain the maximum variance in **X** (like PCA), it determines linear combinations of **X** that have the maximum covariance with **Y**.

PLS2 Algorithm: Outer Relations

- ❖ The **outer relations** for **X** and **Y** are give below

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

- ❖ Dimensions of **T** and **U** are (n x a) where 'a' is the number of factors/components/latent variables
- ❖ Dimensions of **P** and **Q** are (p x a) and (k x a) respectively
- ❖ Dimensions of **E** and **F*** are (n x p) and (n x k) respectively

PLS2 Algorithm: Iterations

- ❖ During the iterative part of this algorithm, the result from $\mathbf{X}$ is passed over for finding the value of $\mathbf{Y}$ and vice versa
- ❖ The **outer relations** for $\mathbf{X}$ and $\mathbf{Y}$ are give below ($\mathbf{X}$ and $\mathbf{Y}$ are decomposed)

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

- ❖ The **inner relation** *between* $\mathbf{X}$ and $\mathbf{Y}$ involving $\mathbf{T}$ and $\mathbf{U}$ is given as follows

$$\mathbf{U} = \mathbf{T}\mathbf{B} + \mathbf{H}$$

Thus, we relate $\mathbf{Y}$ with $\mathbf{T}$

- ❖ $\mathbf{B}$ is a diagonal matrix of dimension $(a \times a)$ and $\mathbf{H}$ has dimensions of $(n \times a)$

PLS2: Decomposing Y

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

Outer Relation

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{U} = \mathbf{T}\mathbf{B} + \mathbf{H}$$

Inner Relation

❖ Hence, from the above we end up with the mixed relation

$$\mathbf{Y} = (\mathbf{T}\mathbf{B} + \mathbf{H})\mathbf{Q}' + \mathbf{F}^*$$

(remember $\mathbf{B}$ is a diagonal matrix)

$$\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + (\mathbf{H}\mathbf{Q}' + \mathbf{F}^*)$$

$$\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F}$$

Mixed Relation

❖ Here, $\mathbf{F}$ refers to the residual matrix of the mixed relation while $\mathbf{F}^*$ and $\mathbf{E}$ refer to the residuals of the outer relations

❖ $\mathbf{E}$ and $\mathbf{F}$ will be used during the iterations of the algorithm

PLS2 Algorithm: Summary of Relations

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

Outer Relation

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{U} = \mathbf{T}\mathbf{B} + \mathbf{H}$$

Inner Relation

$$\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F}$$

Mixed Relation

- ❖ Next the algorithm seeks to **get estimates** for **W, T, P, U, Q** and **B**
- ❖ The steps involved are discussed in the next slides

PLS2 Algorithm

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{U} = \mathbf{T}\mathbf{B} + \mathbf{H}$$

$$\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F}$$

- ❖ Seek estimates for $\mathbf{W}$, $\mathbf{T}$, $\mathbf{P}$, $\mathbf{U}$, $\mathbf{Q}$ and $\mathbf{B}$
- ❖ Begin with $\mathbf{E}_0$ and $\mathbf{F}_0$ represent the mean-centered and scaled (standardized) data matrices $\mathbf{X}$ and $\mathbf{Y}$ respectively
- ❖ $\mathbf{E}_0 = \tilde{\mathbf{X}}$ and $\mathbf{F}_0 = \tilde{\mathbf{Y}}$ (tilde represents standardized – not compulsory)
- ❖ For every component/factor/latent variable t_h where (h = iteration counter)

$$h = 1, 2, \dots, 'a',$$

the following steps are adopted and iterated upon

PLS2 Algorithm: Steps (0-2)

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E} \quad \mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^* \quad \mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F} \quad \mathbf{E}_0 = \tilde{\mathbf{X}} \text{ and } \mathbf{F}_0 = \tilde{\mathbf{Y}}$$

- ❖ Notice that $\mathbf{u}$ is associated with $\mathbf{Y}$ and $\mathbf{t}$ is associated with $\mathbf{X}$
- ❖ **Step 0:** Choose the first column of the matrix $\mathbf{F}_{h-1}$ as a starting value for $\mathbf{u}_h$

You may try choosing any column in $\mathbf{Y}$ say, the first or one which has highest variance

PLS2 Algorithm: Steps (0-2)

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E} \quad \mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^* \quad \mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F} \quad \mathbf{E}_0 = \tilde{\mathbf{X}} \text{ and } \mathbf{F}_0 = \tilde{\mathbf{Y}}$$

❖ **Step 1:** Perform the following regression using **Y score vectors** to find X weights $\mathbf{w}_h$

$$\mathbf{E}_{h-1} = \mathbf{u}_h \mathbf{w}_h' + \varepsilon_1$$

(**X variation** is mapped to **Y score**)

yielding the least squares solution

$$\mathbf{w}_h' = [\mathbf{u}_h' \mathbf{u}_h]^{-1} \mathbf{u}_h' \mathbf{E}_{h-1}$$

❖ **Step 2:** Normalize $\mathbf{w}_h'$ as shown below (**Why normalize? Assignment 2 Part 1**)

$$\mathbf{w}_h' = \frac{\mathbf{w}_h'}{\|\mathbf{w}_h\|}$$

PLS2 Algorithm: Steps (0-2)

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E} \quad \mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^* \quad \mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F} \quad \mathbf{E}_0 = \tilde{\mathbf{X}} \text{ and } \mathbf{F}_0 = \tilde{\mathbf{Y}}$$

❖ **Step 1:** Perform the following regression using **Y score vectors** to find X weights $\mathbf{w}_h$

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(**X variation** is mapped to **Y score**)

yielding the least squares solution

$$\mathbf{w}_h' = [\mathbf{u}_h' \mathbf{u}_h]^{-1} \mathbf{u}_h' \mathbf{E}_{h-1}$$

❖ **Step 2:** Normalize $\mathbf{w}_h'$ as shown below (**Why normalize? Assignment 2 Part 1**)

$$\mathbf{w}_h' = \frac{\mathbf{w}_h'}{\|\mathbf{w}_h\|}$$

Using U to find W: Justification

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E} \quad \mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

In PLS2 using NIPALS, the $\mathbf{X}$ residual matrix is related to $\mathbf{u}$ (the $\mathbf{Y}$ block score) because the core objective of PLS is to **maximize the covariance** between the predictor variables ($\mathbf{X}$) and the response variables ($\mathbf{Y}$).

$$\mathbf{E}_{h-1} = \mathbf{u}_h \mathbf{w}_h' + \varepsilon_1$$

Relating the $\mathbf{X}$ residuals to $\mathbf{u}$ acts as an iterative bridge to achieve this:

- **Finding the $\mathbf{X}$ -Weights ($\mathbf{w}$):** To find the direction in $\mathbf{X}$ that correlates best with $\mathbf{Y}$, the $\mathbf{X}$ residual matrix is regressed against $\mathbf{u}$ (the summary of $\mathbf{Y}$) using the formula $\mathbf{w} \propto \mathbf{X}_{residual}^T \mathbf{u}$. This weights the original variables based directly on their ability to explain the $\mathbf{Y}$ latent space.

$$\mathbf{w}_h' = [\mathbf{u}_h' \mathbf{u}_h]^{-1} \mathbf{u}_h' \mathbf{E}_{h-1}$$

PLS2 Algorithm: Explanation

- ❖ Imagine $\mathbf{X}$ contains hundreds of process variables and $\mathbf{Y}$ contains several quality measurements.
- ❖ After removing the first latent factor, there is still residual unexplained variation in $\mathbf{X}$.
- ❖ Some of that variation is related to $\mathbf{Y}$, and some is just noise or irrelevant process variation.
- ❖ By relating the $\mathbf{X}$ residuals to $\mathbf{u}$ (the $\mathbf{Y}$ Score vector: Summary Sheet for $\mathbf{Y}$), PLS focuses only on the part of the remaining $\mathbf{X}$ variation that still helps explain $\mathbf{Y}$

$$\mathbf{E}_{h-1} = \mathbf{u}_h \mathbf{w}_h' + \varepsilon_1$$

- ❖ This is what gives PLS its **predictive power**.
- ❖ The $\mathbf{X}$ residual matrix is related to the $\mathbf{Y}$ score vector $\mathbf{u}$ because PLS extracts new $\mathbf{X}$ directions $\mathbf{w}$ based on their ability to explain the remaining $\mathbf{Y}$ structure, not merely on their ability to explain remaining $\mathbf{X}$ variance.

PLS2 Algorithm: Explanation

Why relate X residuals to u ?

The key idea is that PLS aims to find directions in X that are most predictive of Y.

If we only decomposed X based on its own residual variance (as in PCA), we might extract components that explain X well but have little relationship to Y.

Instead, in NIPALS PLS:

$$\mathbf{E}_{h-1} = \mathbf{u}_h \mathbf{w}_h' + \varepsilon_1$$
$$w \propto E^T u$$

The residual X matrix E is projected onto the Y score vector u . This finds the direction in the remaining X variation that has the strongest relationship with the current Y latent variable.

So:

- **PCA:** next X component maximizes variance in X residuals.
- **PLS:** next X component maximizes covariance between X residuals and Y latent structure.

PLS2 Algorithm: Find X Scores & Y Loadings

find X scores, Why regress $\mathbf{E}'_{h-1}$ with $\mathbf{w}_h \mathbf{t}'_h$? Should it not be $\mathbf{p}_h \mathbf{t}'_h$?

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{U} = \mathbf{T}\mathbf{B} + \mathbf{H}$$

$$\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F}$$

$$\mathbf{E}_{h-1} = \mathbf{t}_h \mathbf{w}'_h + \varepsilon_2$$

$$\mathbf{E}'_{h-1} = \mathbf{w}_h \mathbf{t}'_h + \varepsilon'_2$$

$$\mathbf{t}'_h = [\mathbf{w}_h' \mathbf{w}_h]^{-1} \mathbf{w}'_h \mathbf{E}'_{h-1}$$

Step 4: Then perform the regression to find the **Y** loadings (q vectors) as given

by (using X scores)

$$\mathbf{F}_{h-1} = \mathbf{t}_h \mathbf{q}'_h + \varepsilon_3$$

$$\mathbf{E}_{h-1} = \mathbf{u}_h \mathbf{w}'_h + \varepsilon_1$$

$$\mathbf{q}'_h = [\mathbf{t}_h' \mathbf{t}_h]^{-1} \mathbf{t}_h' \mathbf{F}_{h-1} \text{ and after normalizing}$$

Step 5:

$$\mathbf{q}'_h = \frac{\mathbf{q}'_h}{\|\mathbf{q}_h\|}$$

PLS2 Algorithm: Find X Scores & Y Loadings

find X scores, Why regress $\mathbf{E}'_{h-1}$ with $\mathbf{w}_h \mathbf{t}'_h$? Should it not be $\mathbf{p}_h \mathbf{t}'_h$?

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{U} = \mathbf{T}\mathbf{B} + \mathbf{H}$$

$$\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F}$$

$$\mathbf{E}_{h-1} = \mathbf{t}_h \mathbf{w}'_h + \varepsilon_2$$

$$\mathbf{E}'_{h-1} = \mathbf{w}_h \mathbf{t}'_h + \varepsilon'_2$$

$$\mathbf{t}'_h = [\mathbf{w}_h' \mathbf{w}_h]^{-1} \mathbf{w}'_h \mathbf{E}'_{h-1}$$

Step 4: Then perform the regression to find the **Y** loadings (q vectors) as given

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$$\mathbf{F}_{h-1} = \mathbf{t}_h \mathbf{q}'_h + \varepsilon_3$$

$$\mathbf{E}_{h-1} = \mathbf{u}_h \mathbf{w}'_h + \varepsilon_1$$

$$\mathbf{q}'_h = [\mathbf{t}_h' \mathbf{t}_h]^{-1} \mathbf{t}_h' \mathbf{F}_{h-1} \text{ and after normalizing}$$

Step 5:

$$\mathbf{q}'_h = \frac{\mathbf{q}'_h}{\|\mathbf{q}_h\|}$$

PLS2 Algorithm: Find X Scores & Y Loadings

$$\mathbf{E}_{h-1} = \mathbf{t}_h \mathbf{w}_h' + \varepsilon_2$$

$$\mathbf{E}_{h-1}' = \mathbf{w}_h \mathbf{t}_h' + \varepsilon_2'$$

$$\mathbf{t}_h' = [\mathbf{w}_h' \mathbf{w}_h]^{-1} \mathbf{w}_h' \mathbf{E}_{h-1}'$$

$$\mathbf{X} = \mathbf{T} \mathbf{P}' + \mathbf{E}$$

$$\mathbf{Y} = \mathbf{U} \mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{U} = \mathbf{T} \mathbf{B} + \mathbf{H}$$

$$\mathbf{Y} = \mathbf{T} \mathbf{B} \mathbf{Q}' + \mathbf{F}$$

Assignment 2 Part 1 Question: To find X scores, Why regress $\mathbf{E}_{h-1}'$ with $\mathbf{w}_h \mathbf{t}_h'$? Should it not be $\mathbf{p}_h \mathbf{t}_h'$?

See <https://chatgpt.com/c/6a635396-7810-83ee-9182-8f4aa5877a10>

For Y loading vector Why Use X scores in PLS2 Nipals?

PLS is fundamentally building a regression relationship from **X** to **Y**. Therefore, once a predictive X score t has been obtained, we ask:

"How does Y vary along this X latent direction?"

The answer is given by projecting Y onto t , yielding q .

Geometric interpretation

The loading vector q is the least-squares coefficient when approximating Y by the X score:

$$Y \approx tq^T$$

Rearranging the least-squares solution gives

$$q = (t^T t)^{-1} Y^T t$$

Thus q tells us how each Y variable responds to movement along the latent X component t .

$$\mathbf{F}_{h-1} = \mathbf{t}_h \mathbf{q}_h' + \varepsilon_3$$

$$\mathbf{q}_h' = [\mathbf{t}_h' \mathbf{t}_h]^{-1} \mathbf{t}_h' \mathbf{F}_{h-1}$$

$$\mathbf{q}_h = [\mathbf{t}_h' \mathbf{t}_h]^{-1} \mathbf{F}_{h-1}' \mathbf{t}_h$$

Tips For Remembering

- ❖ Y **loading** vector $\mathbf{u}$ helps find the $\mathbf{X}$ *weights* vector $\mathbf{w}$ because it tells us which $\mathbf{X}$ directions are relevant to $\mathbf{Y}$.
- ❖ $\mathbf{X}$ score vector $\mathbf{t}$ helps find the $\mathbf{Y}$ *loadings* vectors $\mathbf{q}$ because $\mathbf{Y}$ is ultimately modeled and predicted from the $\mathbf{X}$ latent scores.
- ❖ So the algorithm alternates between $\mathbf{Y}$ -guided ($\mathbf{U}$) extraction of $\mathbf{X}$ components and $\mathbf{X}$ -guided ($\mathbf{T}$) reconstruction of $\mathbf{Y}$,
- ❖ Ensures the latent variables maximize covariance while remaining useful for prediction.

PLS2: Steps 6 and 7

- ❖ **Step 6:** Perform the regression to obtain new values of $\mathbf{u}_h$ (refer step 0)

$$\mathbf{F}_{h-1} = \mathbf{u}_h \mathbf{q}_h' + \varepsilon_4$$

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

$$\mathbf{F}_{h-1}' = \mathbf{q}_h \mathbf{u}_h' + \varepsilon_4'$$

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{u}_h' = [\mathbf{q}_h' \mathbf{q}_h]^{-1} \mathbf{q}_h' \mathbf{F}_{h-1}'$$

$$\mathbf{U} = \mathbf{T}\mathbf{B} + \mathbf{H}$$

$$\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F}$$

- ❖ **Step 7:** Check the convergence of $\mathbf{t}_h$ using a suitable convergence criteria.
- ❖ If the new $\mathbf{t}_h$ is much different than the previous one according to this criterion
initiate a new iteration from Step 1 with this new value.
- ❖ Else go to Step 8

PLS2: Steps 8-10

- ❖ **Step 8:** Perform the regression to obtain new values of X loadings $\mathbf{p}_h$

$$\mathbf{E}_{h-1} = \mathbf{t}_h \mathbf{p}_h' + \varepsilon_5$$

$$\mathbf{p}_h' = [\mathbf{t}_h' \mathbf{t}_h]^{-1} \mathbf{t}_h' \mathbf{E}_{h-1}$$

$$\mathbf{X} = \mathbf{T} \mathbf{P}' + \mathbf{E}$$

$$\mathbf{Y} = \mathbf{U} \mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{U} = \mathbf{T} \mathbf{B} + \mathbf{H}$$

$$\mathbf{Y} = \mathbf{T} \mathbf{B} \mathbf{Q}' + \mathbf{F}$$

- ❖ **Step 9:** Fit $\mathbf{t}_h$ to the newly gained $\mathbf{p}_h$ as $\mathbf{t}_h = \mathbf{t}_h \|\mathbf{p}_h\|$

- ❖ **Step 10:** Normalize $\mathbf{p}_h'$ as $\mathbf{p}_h' = \frac{\mathbf{p}_h'}{\|\mathbf{p}_h\|}$

Assignment 2 Part 1 Question: Why is this done? Explain briefly.

PLS2: Steps 11-12

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{U} = \mathbf{T}\mathbf{B} + \mathbf{H}$$

$$\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F}$$

- ❖ **Step 11:** Calculate b_h which is the h^{th} diagonal element of $\mathbf{B}$

by regressing $\mathbf{u}_h = \mathbf{t}_h \mathbf{b}_h + \varepsilon_6$

with the least squares solution given by

$$\mathbf{b}_h = [\mathbf{t}_h' \mathbf{t}_h]^{-1} \mathbf{t}_h' \mathbf{u}_h$$

- ❖ **Step 12:** Update the data matrices as show below.

- ❖ Deflation removes parts of the $\mathbf{X}$ and $\mathbf{Y}$ matrices that have already been explained.

$$\mathbf{E}_h = \mathbf{E}_{h-1} - \mathbf{t}_h \mathbf{p}_h'$$

$$\mathbf{F}_h = \mathbf{F}_{h-1} - \mathbf{t}_h \mathbf{b}_h \mathbf{q}_h'$$

Discussion on the NIPALS algorithm

- ❖ **Step 1:** Regression using **Y score vectors** to find X weights $\mathbf{w}_h$

$$\mathbf{E}_{h-1} = \mathbf{u}_h \mathbf{w}_h' + \varepsilon_1$$

yielding the least squares solution $\mathbf{w}_h' = [\mathbf{u}_h' \mathbf{u}_h]^{-1} \mathbf{u}_h' \mathbf{E}_{h-1}$

- ❖ **Step 2:** Normalize $\mathbf{w}_h'$ as shown below $\mathbf{w}_h' = \frac{\mathbf{w}_h'}{\|\mathbf{w}_h\|}$

- ❖ **Step 3:** Regression to find **X** scores vector $\mathbf{t}_h$

$$\mathbf{E}_{h-1} = \mathbf{t}_h \mathbf{w}_h' + \varepsilon_2 \quad \mathbf{E}_{h-1}' = \mathbf{w}_h \mathbf{t}_h' + \varepsilon_2' \quad \mathbf{t}_h' = [\mathbf{w}_h' \mathbf{w}_h]^{-1} \mathbf{w}_h' \mathbf{E}_{h-1}'$$

Steps 1-3 apply to **X** data matrix

Discussion on the NIPALS algorithm

Step 4: Now perform the regression to find the $\mathbf{Y}$ weights (q vectors) as given by

$$\mathbf{F}_{h-1} = \mathbf{t}_h \mathbf{q}'_h + \varepsilon_3$$

Step 5:

$$\mathbf{q}'_h = \frac{\mathbf{q}'_h}{\|\mathbf{q}_h\|}$$

Step 6: Perform the regression to obtain new values of u_h (refer step 0)

$$\mathbf{F}_{h-1} = \mathbf{u}_h \mathbf{q}'_h + \varepsilon_4 \quad \mathbf{F}'_{h-1} = \mathbf{q}_h \mathbf{u}'_h + \varepsilon'_4 \quad \mathbf{u}'_h = [\mathbf{q}_h' \mathbf{q}_h]^{-1} \mathbf{q}_h' \mathbf{F}'_{h-1}$$

Steps 4-6 apply to $\mathbf{Y}$ data matrix

PLS2: Convergence Criterion

For Step 7 convergence, the criterion is

- ❖ The norm of the difference between the newly calculated $\mathbf{t}_h$ and the one from the previous iteration $\mathbf{t}_{h-1}$ is lower than a prescribed value ε_c , i.e.,

$$\|\mathbf{t}_h - \mathbf{t}_{h-1}\| \leq \varepsilon_c$$

- ❖ After the convergence criterion is met, $\mathbf{p}_h$ are fitted to $\mathbf{t}_h$ in Step 8

$$\begin{aligned}\mathbf{E}_{h-1} &= \mathbf{t}_h \mathbf{p}_h' + \varepsilon_5 \\ \mathbf{p}_h' &= [\mathbf{t}_h' \mathbf{t}_h]^{-1} \mathbf{q}_h' \mathbf{E}_{h-1}\end{aligned}$$

PLS2: Steps 9 and 10

❖ **Step 9:** Fit $\mathbf{t}_h$ to the newly gained $\mathbf{p}_h$ as $\mathbf{t}_h = \mathbf{t}_h \|\mathbf{p}_h\|$

❖ **Step 10:** Normalize $\mathbf{p}'_h$ as $\mathbf{p}'_h = \frac{\mathbf{p}'_h}{\|\mathbf{p}_h\|}$

❖ In Step 10, $\mathbf{p}_h$ are normalized to length 1 to make them comparable to principal components.

❖ The $\mathbf{t}_h$ have to be fitted to the new $\mathbf{p}_h$ in Step 9

PLS2: Steps 11 and 12

❖ **Step 11:** Calculate b_h which is the h^{th} diagonal element of **B** by

regressing $\mathbf{u}_h = \mathbf{t}_h \mathbf{b}_h + \varepsilon_6$

with the least squares solution given by $\mathbf{b}_h = [\mathbf{t}_h' \mathbf{t}_h]^{-1} \mathbf{t}_h' \mathbf{u}_h$

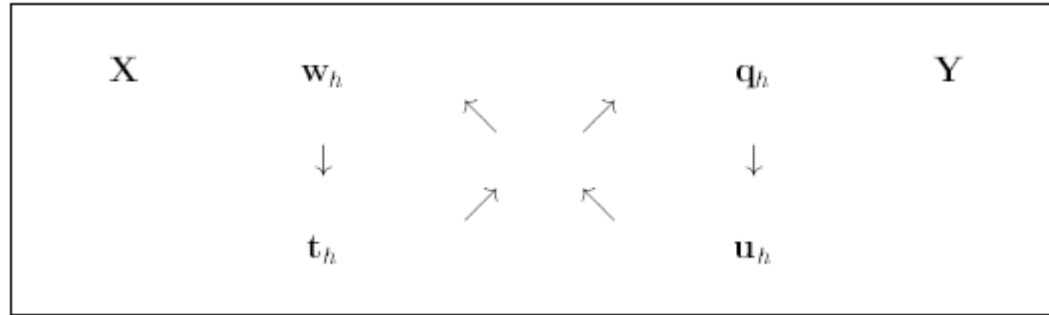
In Step 11, $\mathbf{b}_h$ is found

❖ which is important for updating the data matrix **Y** in Step 12

❖ AND in prediction at a later stage

In Step 12, updates for matrices **X** and **Y** are calculated

PLS2: Graphical Representation



$$X = TP' + E$$

$$Y = UQ' + F^*$$

$$U = TB + H$$

$$Y = TBQ' + F$$

Mathematical flow for one component

$$u_h \xrightarrow{X^T} w_h \xrightarrow{X} t_h \xrightarrow{Y^T} q_h \xrightarrow{Y} u_h$$

This cycle is the essence of the **NIPALS PLS2 algorithm**, seeking latent scores t_h and u_h that maximize the covariance between the predictor block X and the response block Y .

PLS2: Updating the Data Matrices

- ❖ The reason for using $\mathbf{F}$ and $\mathbf{F}^*$ is for predicting values, the goal of PLS.
- ❖ For this, use the mixed relation $\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F}$ instead of the outer relation

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

- ❖ Further, the rank of $\mathbf{Y}$ is not reduced by 1 for each component, so it is really possible to deduce the maximal number of components, namely as many as the rank of $\mathbf{X}$.

$$\mathbf{X} = \mathbf{T}\mathbf{P}' + \mathbf{E}$$

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}' + \mathbf{F}^*$$

$$\mathbf{U} = \mathbf{T}\mathbf{B} + \mathbf{H}$$

$$\mathbf{Y} = \mathbf{T}\mathbf{B}\mathbf{Q}' + \mathbf{F}$$

Steps 1 to 3 represent the NIPALS algorithm applied to the $\mathbf{X}$ -data matrix, whereas in Steps 4 to 6 NIPALS is applied to the $\mathbf{Y}$ -data. Normally, as a criterion for convergence in Step 7, the norm of the difference between the newly calculated $\mathbf{t}_h$ and the one from the previous iteration has to be smaller than a given constant. After this convergence check the $\mathbf{p}_h$ are fitted to the solution $\mathbf{t}_h$ in Step 8.

- ❖ PLS regression **decomposes** both $\mathbf{X}$ and $\mathbf{Y}$ as a product of a common set of **orthogonal factors** and a set of specific loadings
- ❖ The independent variables are **decomposed** as

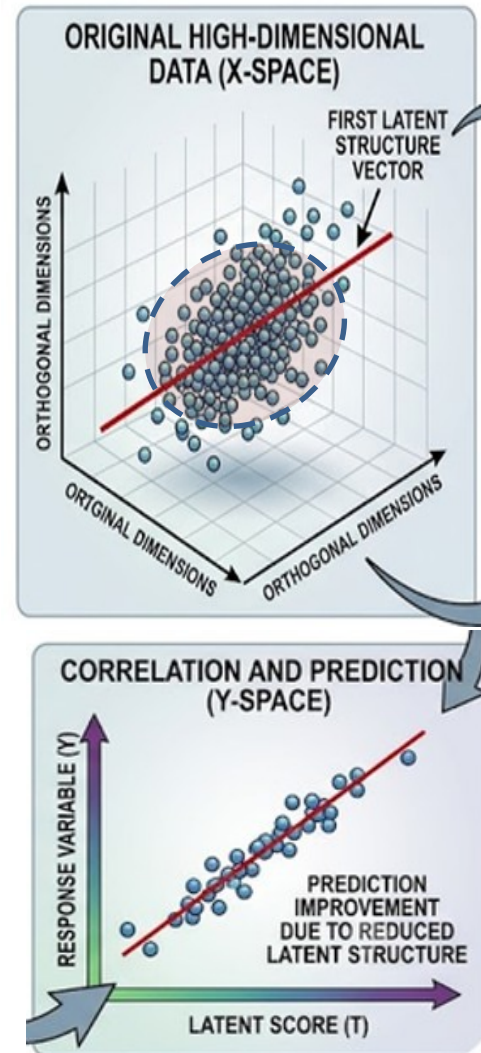
$$\mathbf{X} = \mathbf{TP}'$$

with

$$\mathbf{T}'\mathbf{T} = \mathbf{I}, (\mathbf{T}' = \mathbf{T}^{-1})$$

PLS: Outcomes

- ❖ PLS is a projection method that **reduces** dimensionality.
- ❖ We have taken a **three-dimensional cloud of points** and represented those points using a one-dimensional subspace, namely a line.
- ❖ We say that we have extracted one factor from the data.
- ❖ We have used this one factor to define a linear subspace onto which to project both the Xs and the Ys.



Step 5. Obtain the regression coefficient matrix

Substituting the score equation gives

$$Y = XW(P^TW)^{-1}Q^T$$

Hence the regression coefficient matrix is

$$\beta = W(P^TW)^{-1}Q^T$$

This is the most important equation in PLS2.

The dimensions are

Matrix	Dimensions
X	$n \times p$
Y	$n \times q$
W	$p \times A$
P	$p \times A$
Q	$q \times A$
β	$p \times q$